

Take $a > 0$, we want to evaluate integrals with terms like

$$\sqrt{a^2 + x^2}, \sqrt{x^2 - a^2}, \sqrt{a^2 - x^2}$$

Sometimes this is easy like

$$\int x \sqrt{x^2 - a^2} dx$$

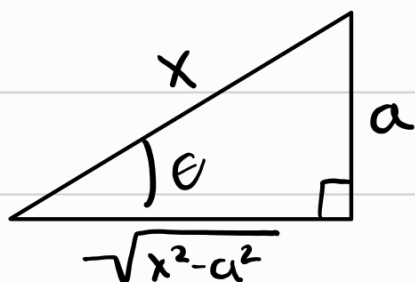
other times, it is not

$$\int \sqrt{x^2 - a^2}$$

let's make a strange substitution

$$x = a \sin \theta$$

this comes from looking at the following triangle



$$\text{in which } \sin \theta = \frac{x}{a}$$

then

$$\begin{aligned} \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2 (1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} \\ &= a \cos \theta \end{aligned}$$

which is much simpler,

to finish the integral, we need dx

$$x = a \sin \theta, \quad dx = a \cos \theta d\theta$$

so

$$\int \sqrt{x^2 - a^2} dx = a^2 \int \cos^2 \theta d\theta \quad \text{which we know.}$$

Notice, usual u -sub involves finding $u(x)$ a function of x .
but now we write $x = f(\theta)$ so x is a function of θ .
So to get back to x we need to inverse function

In general

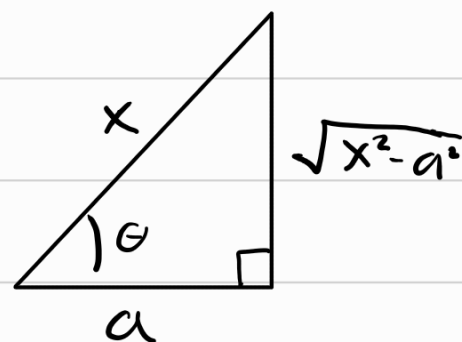
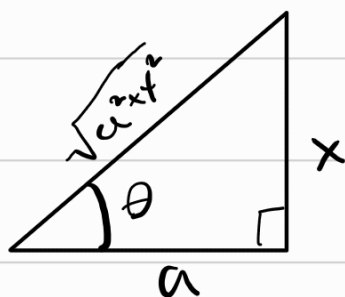
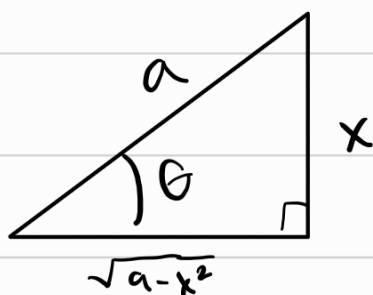
$$\int f(x) dx = \int f(g(t)) g'(t) dt$$

this is called inverse substitution.

Making Substitution

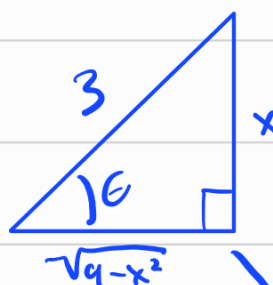
Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

An easier way to see this is via triangles



Ex:

$$\int \frac{\sqrt{4-x^2}}{x^2} dx$$



$$\sim \begin{aligned} x &= 3 \sin \theta \\ dx &= 3 \cos \theta d\theta \end{aligned}$$

$$= \int \frac{3 \cos(\theta)}{(3 \sin \theta)^2} 3 \cos \theta d\theta$$

$$\sqrt{4-x^2} = 3 \cos \theta$$

$$= \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta$$

$$\cot \theta = \frac{\sqrt{4-x^2}}{x}$$

$$= \int \csc^2 \theta - 1 d\theta$$

$$\theta = \arcsin\left(\frac{x}{3}\right)$$

$$= -\cot \theta - \theta + C$$

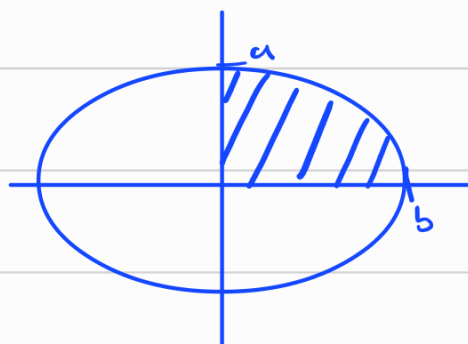
$$= \frac{\sqrt{4-x^2}}{x} - \arcsin\left(\frac{x}{3}\right) + C$$

Ex: Area of Ellipse

Ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2}$

Solving for y ,

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$



We just need quadrant 1, then multiply by 4 to get total Area. So

$$y = \frac{b}{a} \sqrt{a^2 - x^2} \quad 0 \leq x \leq a$$

$$A = 4 \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} dx$$

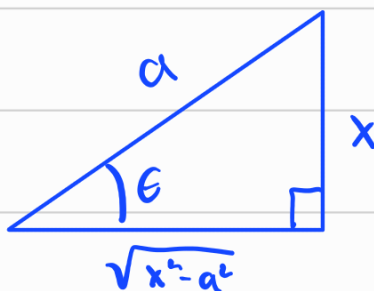
$$= 4 \frac{b}{a} \int_0^{\pi/2} a^2 \cos^2 \theta d\theta$$

$$= 4ab \int_0^{\pi/2} \frac{1}{2} (1 + \cos(2\theta)) d\theta$$

$$= 2ab \left[\theta - \frac{1}{2} \sin(2\theta) \right]_0^{\pi/2}$$

$$= 2ab \left[\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right]$$

$A = ab\pi$



$$x = a \sin \theta$$
$$dx = a \cos \theta d\theta$$

$$\sqrt{x^2 - a^2} = a \cos \theta$$

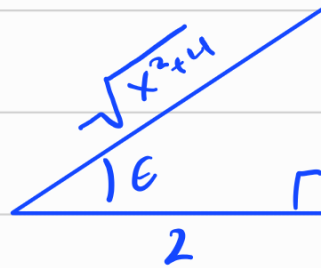
$$\theta = \arcsin\left(\frac{x}{a}\right)$$

$$\theta_1 = \arcsin\left(\frac{a}{a}\right) = \arcsin(1) = \frac{\pi}{2}$$

$$\theta_0 = \arcsin\left(\frac{0}{a}\right) = \arcsin(0) = 0$$

Ex:

$$\int \frac{1}{x^2 \sqrt{x^2+4}} dx$$



$$x = 2 \tan \theta$$
$$dx = 2 \sec^2 \theta d\theta$$

$$\sqrt{x^2+4} = 2 \sec \theta$$

$$\int \frac{2 \sec^2 \theta}{8 \tan^2 \theta \sec \theta} d\theta$$

$$\frac{1/2 \sec \theta}{\tan^2 \theta} = \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta}} = \frac{\cos \theta}{\sin^2 \theta}$$

$$= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta = \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$u = \sin \theta, du = \cos \theta d\theta$$

$$= \frac{1}{4} \int \frac{du}{u^2} = -\frac{1}{4} \cdot \frac{1}{u} + C$$

$$= -\frac{1}{4} \cdot \frac{1}{\sin \theta} + C$$

$$= -\frac{1}{4} \csc \theta + C$$

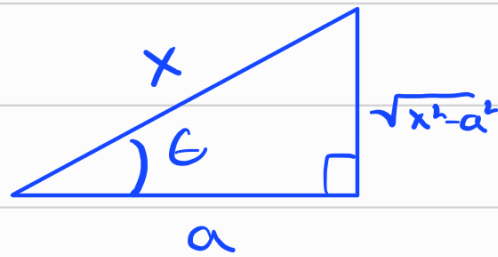
$$= -\frac{\sqrt{x^2+4}}{4x} + C$$

Note: You can, but shouldn't use trig sub

on integrals like $\int \frac{x}{\sqrt{x^2+4}} dx$ as regular u-sub is quicker.

Ex:

$$\int \frac{dx}{\sqrt{x^2 - a^2}}$$



$$x = a \sec \theta$$

$$dx = a \sec \theta \tan \theta d\theta$$

$$\sqrt{x^2 - a^2} = a \tan(\theta)$$

$$= \int \sec \theta d\theta$$

$$= \ln | \sec \theta + \tan \theta | + C$$

$$= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C$$

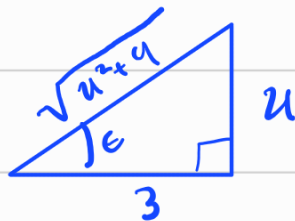
$$= \ln | x + \sqrt{x^2 - a^2} | + \ln(a) + C$$

Ex:

$$\int_0^{3 \cdot \frac{\sqrt{3}}{2}} \frac{x^3}{(\sqrt{4x^2+9})^3} dx$$

$$\text{let } u = 2x \\ du = 2$$

$$\frac{1}{16} \int_0^{3\sqrt{3}} \frac{u^3}{(\sqrt{u^2+9})^3} du$$



$$u = 3 \tan \theta \\ du = 3 \sec^2 \theta d\theta \\ \sqrt{u^2+9} = 3 \sec \theta$$

$$= \frac{1}{16} \int_0^{\pi/3} \frac{27 \tan^3 \theta}{27 \sec^3 \theta} 3 \sec^2 \theta d\theta$$

$$\theta_1 = \arctan \frac{u}{3} = \arctan(\sqrt{3}) = \pi/3$$

$$\theta_0 = \arctan\left(\frac{0}{3}\right) = 0$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta$$

$$\frac{\tan^3 \theta}{\sec \theta} = \frac{\frac{\sin^3 \theta}{\cos^3 \theta}}{\frac{1}{\cos \theta}} = \frac{\sin^3 \theta}{\cos^2 \theta}$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta$$

$$u = \cos \theta \\ du = -\sin \theta d\theta$$

$$= \frac{-3}{16} \int_1^{1/2} \frac{1}{u^2} - 1 du$$

$$= \frac{-3}{16} \left[-\frac{1}{u} - u \right]_1^{1/2}$$

$$= \frac{-3}{16} \left[(-2 - \frac{1}{2}) - (-1 - 1) \right]$$

$$= \frac{-3}{16} \left[-\frac{5}{2} + 2 \right]$$

$$= \frac{-3}{16} \left(-\frac{1}{2} \right) = \boxed{\frac{3}{32}}$$

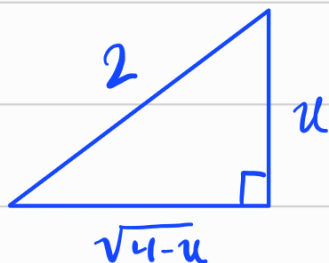
Ex:

$$\int \frac{x}{\sqrt{3-2x-x^2}} dx$$

$$3-2x-x^2 = 3+1-1-2x-x^2 = 4-(x+1)^2$$

$$= \int \frac{x}{\sqrt{4-(x+1)^2}} dx, \quad u = x+1 \rightarrow x = u-1 \\ du = dx$$

$$= \int \frac{u-1}{\sqrt{4-u^2}} du$$



$$= \int \frac{2\sin\theta - 1}{2\cos\theta} 2\cos\theta d\theta$$

$$= \int 2\sin\theta - 1 d\theta$$

$$= -2\cos\theta - \theta + C$$

$$= -\sqrt{4-u^2} - \arcsin\left(\frac{u}{2}\right) + C$$

$$= -\sqrt{4-(x+1)^2} - \arcsin\left(\frac{x+1}{2}\right) + C$$

$$\boxed{= -\sqrt{3-2x-x^2} - \arcsin\left(\frac{x+1}{2}\right) + C}$$

$$u = 2\sin\theta$$

$$du = 2\cos\theta d\theta$$

$$\sqrt{4-u^2} = 2\cos\theta$$